

Output-Controlled TD

Status: independent main proposal with completed 20-seed extended evidence. The current result path is `experiments/alberta_core_rl/results/output_controlled_td/20260709T051934Z_extended`. The CPU task `core-rl-output-extended-fixed-46602102` succeeded on 2026-07-09 and produced standard artifacts. Because the rjob-owned result directory is not writable from the normal project shell, the report-ready figures are stored in this report directory under `figures/`.

Abstract

This proposal studies a practical instability in streaming temporal-difference learning: a fixed parameter step size does not correspond to a fixed prediction change. In linear function approximation, multiplying features by a constant can leave the represented prediction problem conceptually unchanged while changing the effect of the same alpha by orders of magnitude. We test fixed TD, normalized TD, trace-normalized TD(lambda), and a raw-alpha true-online TD(lambda) baseline in a tile-coded random-walk prediction task with feature-scale stress. The extended 20-seed result shows a sharp stability split: normalized and trace-normalized TD have zero seed-level divergence across 400 seed-conditions each, while fixed TD and the current raw-alpha true-online baseline each diverge in 141/400 seed-conditions. The core claim is not that one alpha wins a benchmark, but that online TD needs update units tied to prediction/output effects rather than raw parameter displacement.

Proposal Template Answers

What do we want to understand? We want to understand whether online TD prediction can be made robust to nuisance feature scaling when the learner receives one transition, updates once, and discards the transition.

What setting or testbed is used? The testbed is a tile-coded random-walk prediction problem. The underlying Markov chain and target value function are fixed, but the feature magnitudes are rescaled by uniform, uneven, and lognormal patterns.

What will we examine? We compare fixed-step TD, normalized TD, trace-normalized TD(lambda), and true-online TD(lambda) under a grid of feature scales and alphas. The comparison is deliberately about stability regions, not only final RMSE under one tuned alpha.

What will we look at? The main evidence is a set of heatmaps and tables for RMSE, seed-level divergence, weight norms, prediction-change magnitude, and effective step size. A method is stronger if the same alpha range remains stable across feature scales.

Independent Research Scope

This proposal is an independent study of update units in streaming value prediction. It isolates the feature-scale side of the broader unit-invariance problem: the underlying random-walk prediction problem is kept conceptually fixed while the feature vector is rescaled. Reward-origin effects, control behavior, and model-based planning are intentionally outside this report's scope and are handled by separate proposals.

The proposal should also not be read as a general rejection of true-online TD(lambda). The current true-online condition is a raw-alpha trace baseline in a feature-scale stress test. Its divergence is evidence that raw parameter-space

alpha is unfair under feature rescaling, not evidence that true-online TD(λ) is intrinsically unstable when tuned or normalized appropriately.

Evidence Level

Evidence level: strong independent main-candidate evidence for the prediction-side mechanism, with a fairness caveat for trace baselines. The completed CPU run uses 20 seeds, 20000 steps, five feature-scale patterns, four alphas, and four algorithms. The key result is seed-level rather than row-level: normalized and trace-normalized TD have zero divergent seed-conditions across the full grid, while fixed TD and the raw-alpha true-online baseline diverge in 141 of 400 seed-conditions each.

The evidence is not yet a full control result and not yet a complete true-online TD(λ) audit. The next experiment should add max-stable-alpha tables, normalized true-online TD(λ), and a no-reset feature-scale switch in a single stream. Those follow-ups are needed before the report can claim that output-controlled updates solve feature-unit drift in control agents.

Research Motivation

Streaming RL removes the stabilizing effects of replay buffers, minibatches, and repeated passes over data. Every update is made from the current transition. If a feature vector is unusually large, a parameter-space alpha can produce a huge prediction change before the learner sees any corrective sample. This is not a cosmetic scaling issue: a long-lived agent in the Alberta Plan sense may maintain many value functions, GVFs, and learned models over sensory channels whose units drift or differ by construction.

The Alberta Plan emphasizes ordinary experience, temporal uniformity, continual value-function learning, and limited computation. Those commitments make manual per-sensor alpha tuning unattractive. A more scalable principle is to define the size of an update by its intended effect on the prediction, echoing normalized LMS and recent intentional-update arguments for streaming RL. This proposal tests that idea in the simplest setting where feature scale and prediction error can be inspected exactly.

Research Question

Can output-controlled or normalized TD make streaming value prediction robust to feature scale and trace magnitude?

Hypothesis: normalized TD variants should have a larger stable region over feature scales and step sizes than fixed-step TD, because their alpha approximately controls an output change rather than a raw parameter movement.

Alberta Plan Connection

This study connects to the value-functions component of the Alberta Plan base agent. A continual agent may learn many predictions from the same stream, and those predictions must be updated every time step with limited computation. The proposal also connects to temporal uniformity: there is no special calibration phase, no replay buffer, and no retrospective rescaling of the dataset. The agent must learn as experience arrives.

Related Work

The closest recent motivation is Intentional Updates for Streaming Reinforcement

Learning, which argues that parameter-space step sizes can produce unpredictable output changes in batch-size-one learning and proposes specifying the intended outcome of an update first. Reward Centering motivates a parallel invariance question on the reward side: arbitrary reward offsets should not change continuing-control behavior. Streaming Deep RL Finally Works and Squeezing More from the Stream define the broader no-replay streaming regime, though this project intentionally stays with linear Core RL. True Online TD(λ) is included as an important online trace baseline; the current raw-alpha implementation must be interpreted carefully because true-online methods were designed to be strong and stable when step sizes are used appropriately.

Environment

The environment is a larger random-walk prediction task with overlapping tile-coded features. The agent predicts the probability of reaching the right terminal state under a random policy. The true values are known, so RMSE can be measured over all states during the stream.

The feature-scale conditions are one, ten, hundred, uneven, and lognormal. These conditions keep the qualitative prediction task fixed while changing the geometry of the feature vectors. The hundred condition is an intentionally harsh uniform rescaling; uneven and lognormal make some dimensions much larger than others, which is closer to a multi-sensor setting.

Methods

Fixed TD uses a constant alpha and the update direction x_t or z_t . Normalized TD divides alpha by approximately $\epsilon + ||x_t||^2$, so a larger feature vector does not automatically imply a larger prediction movement. Trace-normalized TD(λ) divides by the trace norm because the eligibility trace, not the current feature alone, is the actual update direction when λ is nonzero. True-online TD(λ) is included with λ 0.8 as a principled trace method, but in this report it is a raw-alpha baseline rather than a tuned or normalized true-online variant.

The logged prediction_change metric estimates how much the current prediction moved after the update. This metric is central: an update rule can look reasonable in parameter space while being unstable in prediction space.

Experimental Design

The completed extended sweep uses:

- - result path:
experiments/alberta_core_rl/results/output_controlled_td/20260709T051934Z_extended;
- - CPU task: core-rl-output-extended-fixed-46602102;
- - seeds: 0-19;
- - steps per seed-condition: 20000;
- - representation: tile coding;
- - feature scales: one, ten, hundred, uneven, lognormal;
- - alphas: 0.01, 0.03, 0.1, 0.3;
- - algorithms: fixed TD, normalized TD, trace-normalized TD(λ), raw-alpha true-online TD(λ);
- - λ : 0.8 for trace-normalized and true-online variants, 0 otherwise.

The decision rule is not simply lowest RMSE. A streaming-compatible update is expected to preserve stability across nuisance feature-unit changes. Divergence is therefore evaluated as a seed-level event: a seed-condition counts as divergent if

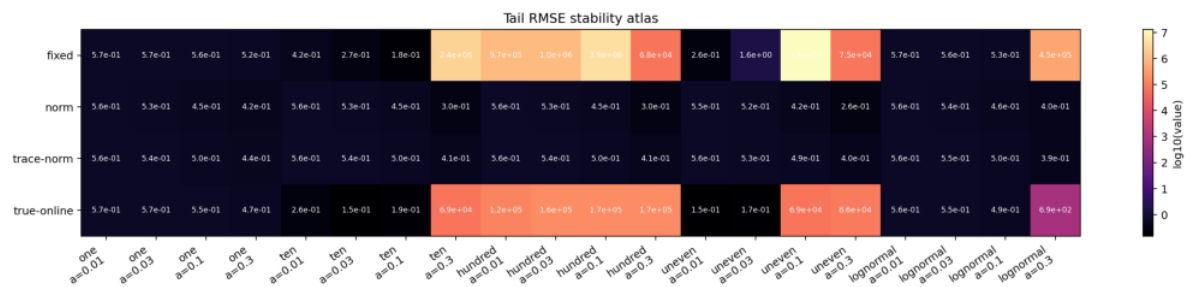
any logged row for that seed-condition has diverged = 1.

Experiment Design Rationale

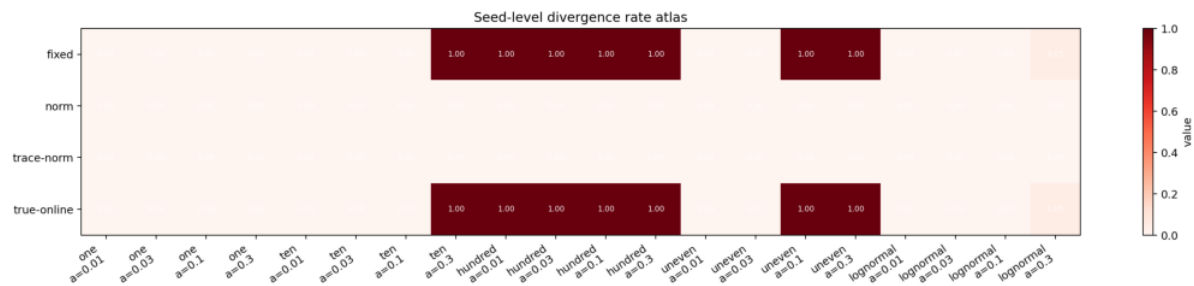
The random-walk prediction task is intentionally simpler than a control benchmark because the research question is about update geometry. A known value function lets the report measure RMSE directly, while tile coding supplies overlapping linear features where feature norms and trace norms matter. This makes it possible to distinguish three mechanisms: prediction error, raw weight movement, and actual output change.

The scale conditions are not meant to be realistic sensor models by themselves. They are invariance tests. Uniform scaling asks whether the same represented prediction can be learned under a different unit. Uneven and lognormal scaling ask whether a few large components can dominate a streaming update. The alpha grid is chosen to reveal stability boundaries, not to crown one tuned setting. The correct figure is therefore a stability atlas and divergence table, not a single final-RMSE leaderboard.

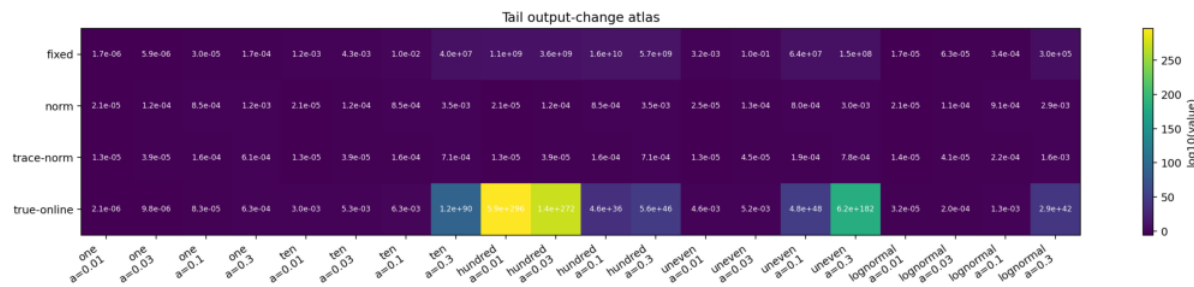
Results



Tail RMSE stability atlas by algorithm, scale, and alpha.



Seed-level divergence-rate atlas by algorithm, scale, and alpha.



Tail output-change atlas by algorithm, scale, and alpha.

Algorithm	Lambda	Seed-conditions diverged	Non-diverged final RMSE mean	Interpretation
fixed TD	0.0	'141/400' ('35.2%')	'0.551'	Stable in easy scales but catastrophically scale-sensitive.

normalized TD	0.0	`0/400` (`0.0%`)	`0.461`	Stable across every tested scale and alpha.
trace-normalized TD(λ)	0.8	`0/400` (`0.0%`)	`0.497`	Stable with traces when the trace direction is normalized.
true-online TD(λ), raw alpha	0.8	`141/400` (`35.2%`)	`0.382`	Low RMSE on surviving easy conditions, but many high-scale conditions diverge.

Fixed TD diverges for all seeds at scale hundred for every tested alpha, at scale ten for alpha 0.3, and at scale uneven for alphas 0.1 and 0.3. It also has a small but real divergence rate under lognormal at alpha 0.3.

Normalized TD has no seed-level divergence across the full grid. At scale hundred, its final RMSE is approximately 0.56, 0.53, 0.44, and 0.28 for alphas 0.01, 0.03, 0.1, and 0.3. Trace-normalized TD(λ) also has no divergence; at scale hundred, its final RMSE is approximately 0.56, 0.54, 0.49, and 0.40.

The raw-alpha true-online TD(λ) baseline should be read carefully. It diverges in the same number of seed-conditions as fixed TD, including all hundred conditions, yet its non-diverged RMSE is low because the surviving conditions are the easier scale settings. This makes it a warning about fair step-size comparison, not a claim that true-online TD(λ) is generally poor.

Analysis

The heatmaps support the central mechanism: fixed alpha is a poor unit for streaming prediction progress when feature scale changes. Under large or uneven scales, the same TD error can create a much larger output change simply because the feature vector is larger. Normalized TD changes the denominator of the update so the effective step size contracts when the feature direction is large. Trace-normalized TD applies the same idea to the accumulated eligibility trace.

The result also clarifies why a one-alpha leaderboard would be misleading. True-online TD(λ) has strong theory and prior empirical support, but a raw-alpha comparison against normalized methods is not fair when the experimental manipulation is feature magnitude. The next version should include a max-stable-alpha audit and a normalized true-online variant before using true-online TD(λ) as a negative baseline.

Threats To Validity

The task is prediction-only. That makes the feature-scale mechanism clean, but a full control study is needed to test whether output-controlled updates help Sarsa or actor-critic learning.

Feature scaling is synthetic. This is appropriate for an invariance test, but real sensor streams may have drifting scale, changing relevance, and partial observability. A no-reset feature-scale switch is the most important next experiment.

The divergence metric is now seed-level, which is more interpretable than logged-row averages, but final RMSE is still measured at the last logged point rather than as an area-under-learning-curve statistic. Future analysis should report both stability and learning speed.

Reviewer Critique And Revisions

Strict Core-RL reviewer: a random-walk prediction task can look toy-like. Response: the final experiment uses overlapping tile coding, five feature-scale patterns, 20 seeds, and a 4-by-5 scale/alpha stress grid. The environment is simple because the question is about update geometry, not behavior complexity.

Function-approximation reviewer: true-online TD(λ) should not be treated

unfairly. Response: the report now labels it as a raw-alpha baseline and keeps the required next action as a fair true-online audit.

Reproducibility reviewer: the CPU job looked stuck because the old runner wrote results only at the end. Response: the job succeeded; future Output-Controlled TD runs now print condition-level progress. The report also records the rjob ownership issue and stores figures in the report folder.

Per-proposal audit matrix:

```
| Reviewer angle | Critique | Action taken | Remaining risk |
|---|---|---|---|
| Core RL | Random walk may look too simple. | Uses tile coding, feature-scale stress, seed-level divergence, and known-value RMSE to isolate the update-unit mechanism. | Control transfer still needs a Sarsa or actor-critic experiment. |
| Function approximation | Raw-alpha true-online TD(lambda) is not a fair final baseline. | Labels it as raw-alpha and avoids broad claims against true-online TD(lambda). | Needs max-stable-alpha and normalized true-online variants. |
| Streaming learning | A fixed scale per run is weaker than sensor drift. | Frames current result as fixed-condition invariance. | Needs no-reset feature-scale switch. |
| Statistics | Divergence row counts can exaggerate long runs. | Uses seed-level divergence events. | Time-to-divergence and AUC remain useful additions. |
| Strict instructor | The report should answer a question, not say normalized TD wins. | Main claim is about output-controlled update units preserving stability under nuisance feature scaling. | Needs tighter connection to downstream control for final project selection. |
```

Conclusion

The extended evidence supports the proposal's main claim: streaming TD should control update consequences in prediction space, not only raw parameter movement. Normalized TD and trace-normalized TD(lambda) remain stable across all tested feature scales and alphas, while fixed TD is fragile under scale stress. The proposal is strong as an independent Core RL study and also motivates the broader Scale-Invariant Continuing Control proposal.

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/output_controlled_td/config_extended.json
```

CPU-task command used for the current extended evidence:

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PARTITION=safethm_cpu_task CPU=8 MEM=16000 \
bash experiments/alberta_core_rl/scripts/run_cpu_task.sh \
core-rl-output-extended-fixed \
"PYTHONNOUSERSITE=1 python experiments/alberta_core_rl/scripts/run_experiment.py --config
experiments/alberta_core_rl/configs/output_controlled_td/config_extended.json"
```

Report figures were generated into the report folder because the rjob result directory is not writable:

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
python experiments/alberta_core_rl/scripts/plot_report_figures.py \
--kind output-td \
--result-dir experiments/alberta_core_rl/results/output_controlled_td/20260709T051934Z_extended \
--figure-dir final/reports/proposals/output_controlled_td/figures
```